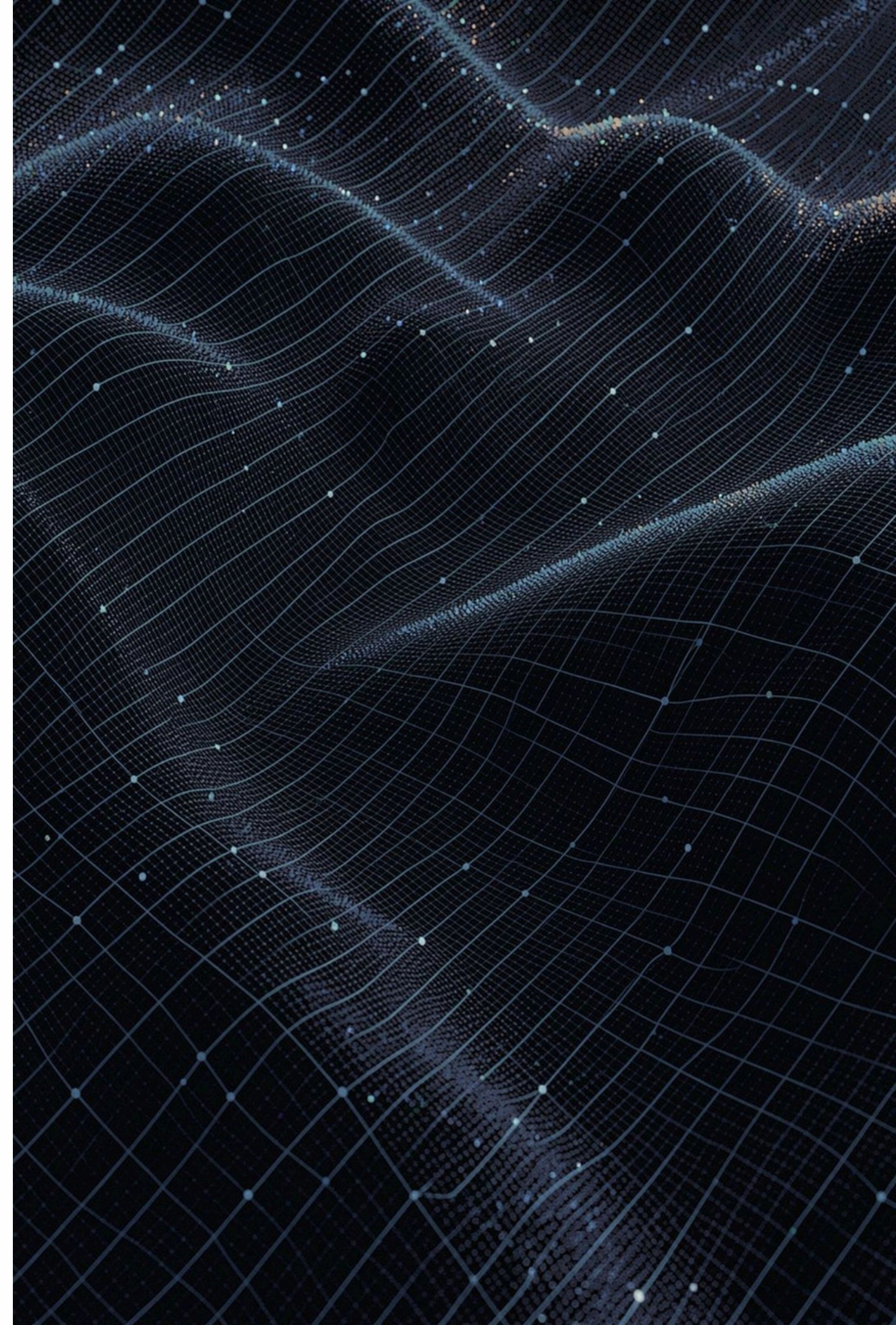


James Andersen

Information Systems | Data Analytics | Business Intelligence | Big Data

Portfolio 2026

jamesandersen112@gmail.com · +62 851-7242-5514 · [linkedin.com/in/james-andersen-06261a270](https://www.linkedin.com/in/james-andersen-06261a270) · github.com/jmsand12



About Me

Information Systems graduate from Multimedia Nusantara University with expertise in **Data Analytics**, **Business Intelligence**, and **Big Data**. Proficient in SQL, Python, Power BI, and Pyspark. Experienced in building data pipelines, analytical dashboards, machine learning models, and information systems that drive business insights.

Data Analytics

Python · SQL · Data
Visualization · Pandas · NumPy

Business Intelligence

Power BI · Power Query · DAX ·
Tableau

Big Data & ML

PySpark · Gradient Boosted Tree · SHAP · RAG



Experience

AUG 2025 – DEC 2025

Web Developer

UMN Student Development

- Developed internal ticketing system for Student Engagement with dashboard KPI and trend visualizations
- Implemented using PHP, PDO, MySQL, JavaScript, Chart.js, and XAMPP
- Participated in full SDLC: planning, analysis, design, implementation, testing, deployment, and maintenance

FEB 2025 – JUN 2025

Junior Data Scientist

VINIX7

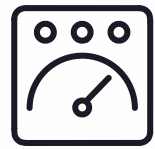
- Apply data acquisition, cleaning, and exploratory data analysis (EDA) techniques using Excel and other tools.
- Create interactive data visualizations and compile business insights using Business Intelligence (BI) principles.
- Work on real case studies as part of the final project, integrating all data science learning materials.
- Produce effective data-driven infographics and storytelling to communicate insights to stakeholders.

Technical Skills



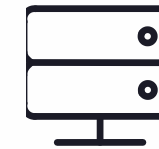
Data Analytics

Python · Pandas · NumPy · Matplotlib ·
SQL · Statistics



Business Intelligence

Power BI · DAX · Power Query · Tableau



Database & Big Data

PostgreSQL · MySQL · PySpark · Spark
SQL · JDBC



Machine Learning & AI

Scikit-learn · Gradient Boosted Tree ·
SHAP · RAG · LLM



Development & Tools

PHP · Git · Jupyter Notebook · VS Code ·
Figma

Customer Behavior Analysis

BUSINESS INTELLIGENCE

DATA ANALYTICS



Objective

Analyze customer shopping behavior to identify purchasing patterns, product preferences, discount impact, subscription opportunities, and customer segments, generating actionable insights to support marketing, retention, and product management decisions.

Approach

Performed end-to-end data analysis by cleaning and preprocessing the dataset with Python, engineering customer-related features, loading the processed data into PostgreSQL, conducting business analysis using SQL, and visualizing the results through an interactive dashboard.

Tools

POWER BI

PYTHON

PostgreSQL

SQL

FraudShield

Large-Scale Digital Transaction Fraud Detection

7.48M+

Transactions

Dataset rows processed at scale

99.3%

ROC-AUC

Model discrimination performance

17K+

Rows/Second

Distributed Spark processing throughput

96.3%

F1-Score

Precision and Recall on Imbalanced Data

Implemented large-scale fraud detection workflow using **PySpark** on a transaction dataset with 7.48M rows. Includes data ingestion, preprocessing, feature engineering, machine learning classification, and explainability analysis.

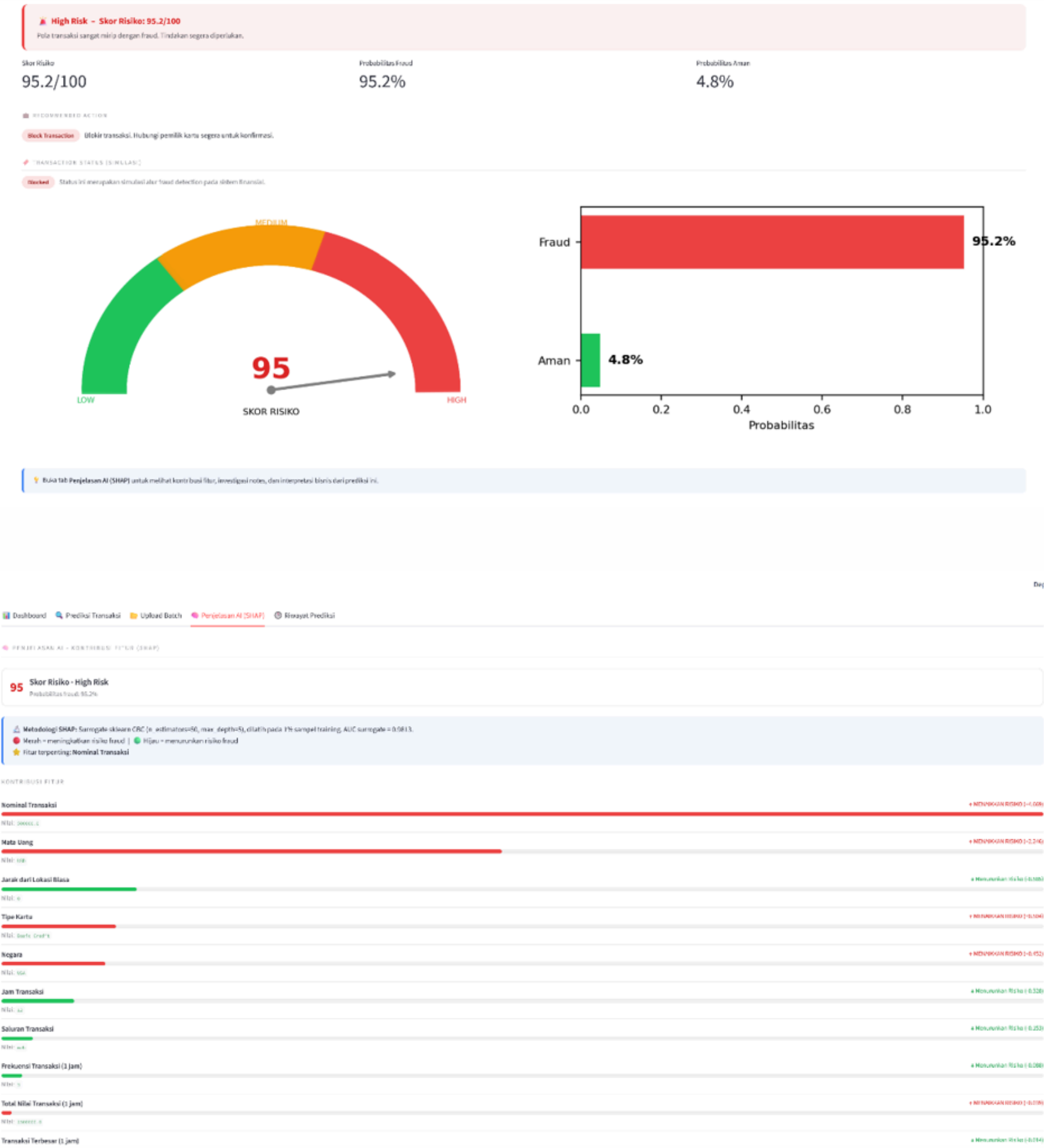
Tools

APACHE SPARK

SHAP

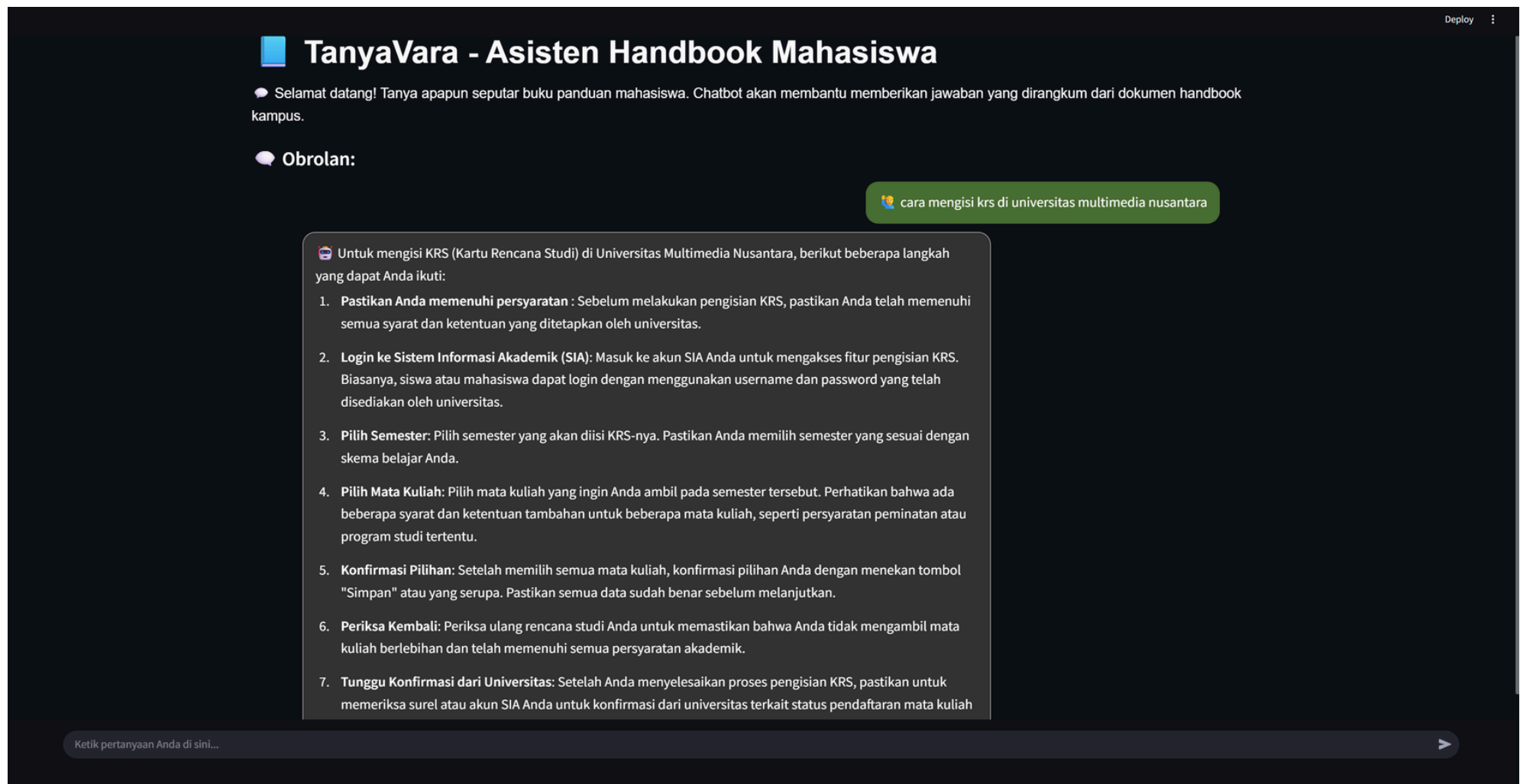
POSTGRESQL

Gradient Boosted Tree



TANYAVARA

RAG-Based University Information Chatbot



Architecture

TANYAVARA uses a **Retrieval-Augmented Generation** architecture to answer university-related queries with grounded, document-backed responses that eliminating hallucination and improving factual accuracy.

Technologies

- RAG
- SENTENCE TRANSFORMERS
- PYMUPDF
- MARIANMT

LLaMA 3.1 8B
Open-weight LLM for response generation

BM25 + Embeddings
Hybrid retrieval for high-recall document search

Streamlit
Lightweight interactive interface for end users

Internal Ticketing System

WEB DEVELOPMENT

INFORMATION SYSTEMS

Objective

Develop an internal ticketing system for **UMN Student Service, Student Support, and Student Development** to streamline request management and provide real-time operational visibility through dashboards.

Implementation

- Dashboard with KPI metrics and trend visualizations using Chart.js
- Full SDLC participation: planning, design, testing, deployment, and maintenance
- Black-box testing for quality assurance

Tools

PHP

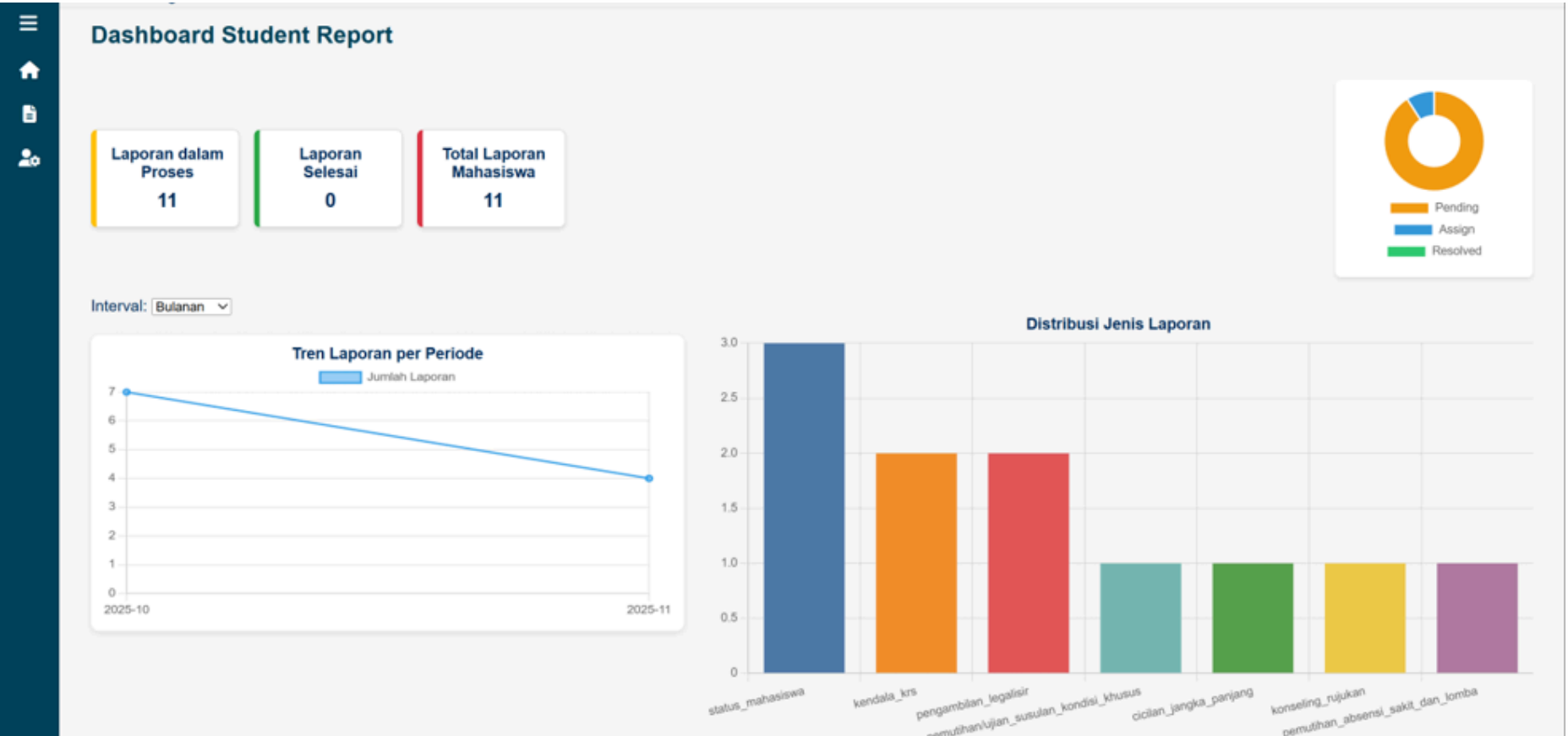
PDO

MYSQL

JAVASCRIPT

CHART.JS

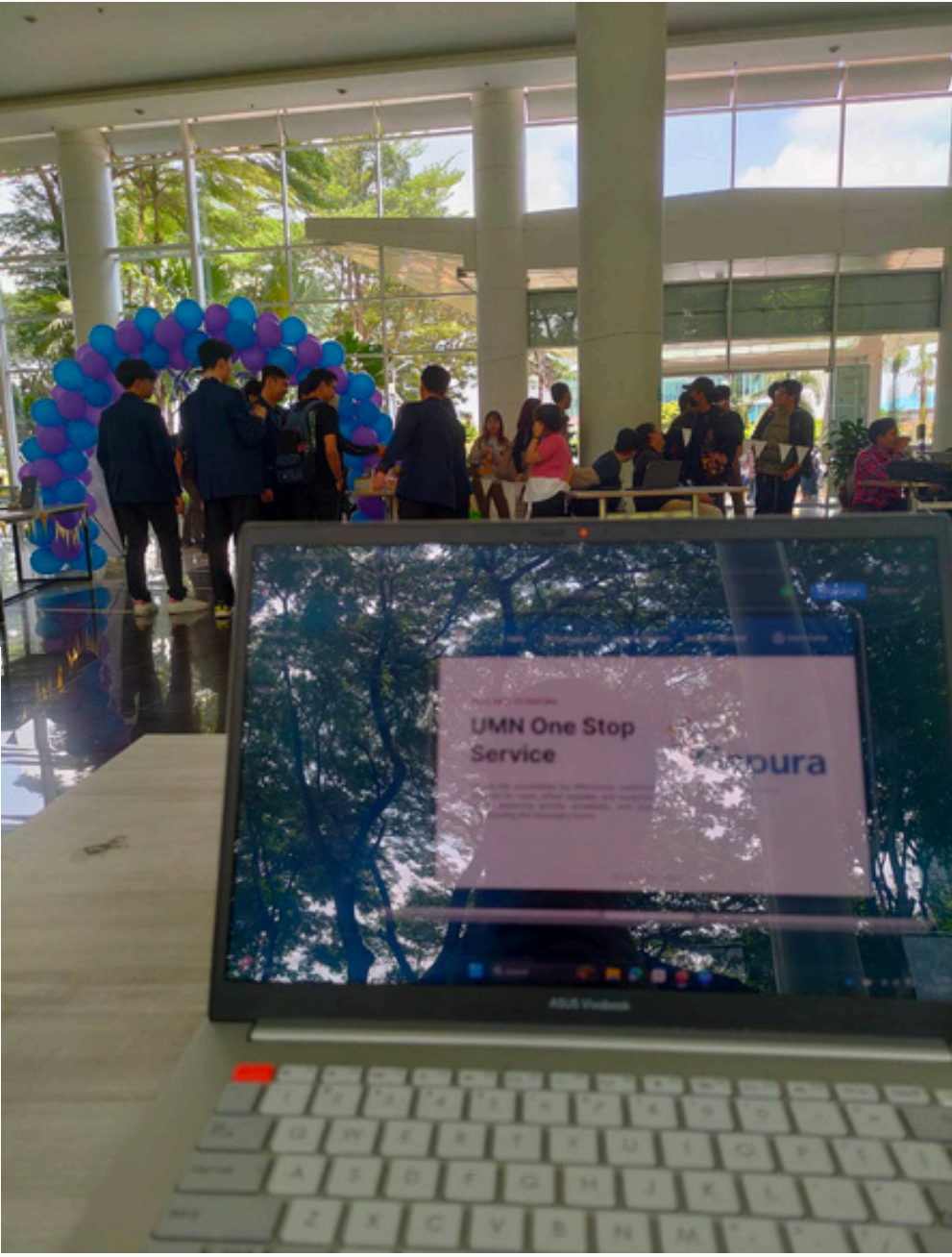
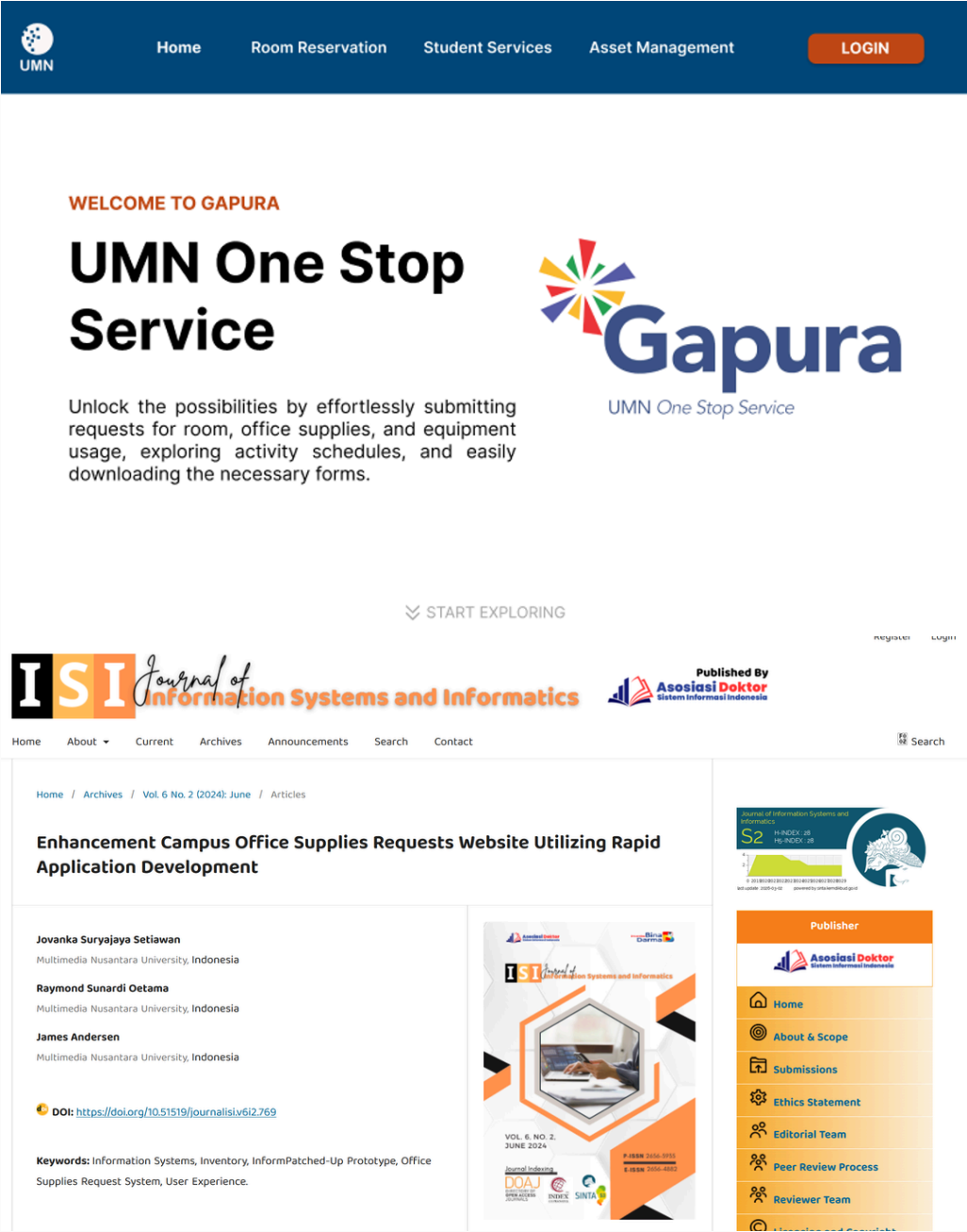
XAMPP



Action	Timestamp	NIM	Nama Mahasiswa	Report type	Forward to	Status
EditDelete	2025-11-07 16:20:10	00000054321	Anderson	kendala_krs	-	pending
EditDelete	2025-11-07 16:19:58	00000069612	James Andersen	status_mahasiswa	-	pending
EditDelete	2025-11-07 16:19:45	00000069612	James Andersen	pengambilan_legalisir	-	pending
EditDelete	2025-11-07 15:36:56	00000069612	James Andersen	status_mahasiswa	Service	pending
EditDelete	2025-10-29 14:21:30	00000069612	James Andersen	status_mahasiswa	Service	pending
EditDelete	2025-10-29 14:19:09	00000054321	Anderson	pemutihan/ujian_susulan_kondisi_khusus	-	pending
EditDelete	2025-10-29 14:18:58	00000012345	Dummy	pengambilan_legalisir	-	pending
EditDelete	2025-10-29 14:18:49	00000054321	Anderson	cicilan_jangka_panjang	-	pending
EditDelete	2025-10-29 14:18:38	00000012345	Dummy	kendala_krs	-	pending
EditDelete	2025-10-29 14:09:09	00000012345	Dummy	konseling_rujukan	Service	assign

Campus Office Supplies Request

- WEB DEVELOPMENT
- INFORMATION SYSTEMS
- UI/UX



Objective

Enhance the campus office supplies request system by improving its usability, workflow, and transparency through user feedback, UI/UX design, and RAD-based development. The improvements were evaluated through User Acceptance Testing (UAT) to assess whether the resulting system better addressed user requirements and supported a more effective request process.

Approach

User Feedback & Requirement Identification → UI/UX Design → RAD-Based Development → Feature Evaluation → User Acceptance Testing

Tools

- RAD
- Figma
- UAT

Stroke Risk Classification Analysis

MACHINE LEARNING

DATA ANALYSIS

CLASSIFICATION

Objective

Develop and evaluate an interpretable Decision Tree model for stroke risk classification by improving data quality, selecting relevant features, and applying pruning techniques. The study aimed to identify an effective combination of feature selection and model optimization to improve classification accuracy.

Approach

Data Preprocessing → UI/UX Design → Feature Selection → Modeling → Evaluation → Interpretation

Findings

The pruned Decision Tree achieved 94.78% accuracy, outperforming the previous Decision Tree (92%) and the evaluated SVM models. The result indicates that selecting BMI and average glucose level, combined with pruning, produced the strongest classification performance in this study.

Tools

Python

Pandas

SCIKIT-LEARN

Jupyter Notebook

- Hasil Evaluasi dari model pada penelitian ini.

Decision Tree				SVM
Sebelum Pruning		Setelah Pruning		
		Max_depth = 3	Max_depth = 4	
Accuracy	0.9134	0.9478	0.9478	0.9197
Cross-Validation Scores	[0.9207, 0.9176, 0.9175, 0.9331, 0.9206]	[0.9582, 0.9593, 0.9582, 0.9592, 0.9592]	[0.9582, 0.9593, 0.9561, 0.9592, 0.9592]	[0.9582, 0.9593, 0.9592, 0.9592, 0.9363]
Mean Accuracy	0.9219	0.9588	0.9584	0.9545
Standard Deviation	0.0057	0.0005	0.0012	0.0090

Tabel 1. Evaluasi penelitian sekarang

- Hasil Evaluasi dari model pada penelitian sebelumnya.

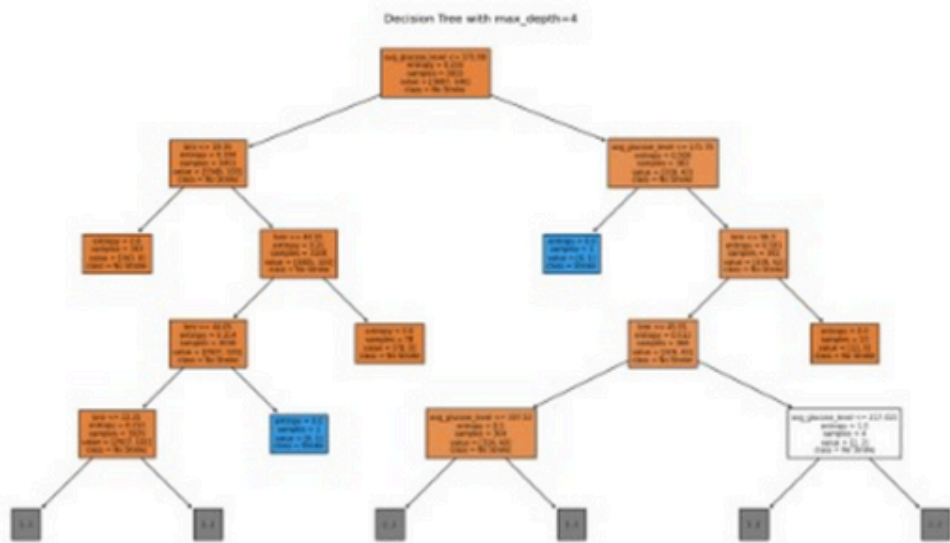
	Decision Tree	SVM (linear)	SVM (polynomial)
Accuracy	0.92	0.76	0/80

Tabel 2. Evaluasi penelitian sebelumnya

- Model decision tree dengan pruning (max_depth = 4)

```
clf_p4 = DecisionTreeClassifier(criterion="entropy", max_depth=4)
clf_p4.fit(X_train, y_train)

y_pred_p = clf_p4.predict(X_test)
```



94.78%

Decision Tree Accuracy

92%

Previous Research Decision Tree

+2.78%

Accuracy Difference

2

Selected Feature (BMI + Average Glucose Level)

Connect with me!

Contact Information



+6285172425514



James Andersen



jamesandersen112@gmail.com